

Equity Quantitative Investment Strategies

From Financial Data to Systematic Strategies

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01/08/2026

Course Description

This course provides a comprehensive introduction to quantitative equity investing. Students will learn how systematic investment strategies are designed, implemented and evaluated following the same research workflow used in quantitative asset management firms.

Rather than presenting statistical and machine learning models in isolation, the course follows the complete quantitative research pipeline, from financial data acquisition to portfolio construction and performance evaluation. Throughout the course, emphasis is placed on developing both the theoretical foundations and the practical implementation skills required to conduct professional quantitative research.

Particular emphasis is placed on mathematical understanding. All major models introduced throughout the course are derived rigorously from first principles before being implemented in Python. In parallel, students progressively build a production-quality quantitative research environment, adopting software engineering practices commonly used in industry, including object-oriented programming, version control with Git and GitHub, containerisation with Docker, cloud computing with Amazon EC2, and cloud storage using Amazon S3.

The course is designed to be accessible to students from business schools, economics and finance programmes, as well as engineering and data science backgrounds. While basic mathematical and programming maturity is expected, all required concepts are introduced progressively throughout the course, allowing students with diverse backgrounds to develop a common quantitative framework.

Beyond the academic content, the course also highlights the concepts and methodologies that are most frequently encountered in quantitative finance interviews, enabling students to connect theoretical knowledge with current industry expectations.

The course combines theoretical lectures with practical Python tutorials and concludes with a research project where students develop, implement and evaluate their own systematic equity investment strategy.

Course Formats

The course is designed to be modular and can be adapted to the needs of different programmes.

- **Introductory Module (12 hours):** *Introduction to Quantitative Equity Investing.* A concise overview of the quantitative research process, covering the main concepts of systematic investing, financial data, signal construction and portfolio evaluation.
- **Semester Course (24–30 hours):** A standard semester-long course introducing the complete quantitative research pipeline, from financial data engineering to portfolio construction and performance analysis, combining theoretical lectures with Python tutorials.

- **Advanced Course (56 hours):** A comprehensive course covering the entire quantitative investment workflow, including production-level Python development (object-oriented programming, Git, GitHub, Docker, Amazon EC2, Amazon S3), statistical and machine learning models, portfolio optimisation, backtesting framework development, and a final research project.

Technical Skills Developed

Throughout the course, students progressively develop a modern quantitative research environment comparable to those used in quantitative asset management firms.

The Python tutorials cover both quantitative finance applications and professional software engineering practices, including

- Python for quantitative finance;
- object-oriented programming;
- software architecture and modular code design;
- Git and GitHub for version control;
- virtual environments and dependency management;
- Docker for reproducible research;
- cloud computing using Amazon EC2;
- cloud storage using Amazon S3;
- reproducible research workflows;
- professional backtesting architecture.

Learning Objectives

Upon successful completion of the course, students will be able to

- understand the complete quantitative equity research process;
- acquire, clean and structure financial datasets;
- engineer predictive financial features;
- derive and implement statistical learning models from first principles;
- construct systematic investment signals;
- build quantitative portfolios under realistic investment constraints;
- design robust backtesting frameworks;
- evaluate investment strategies using industry-standard performance metrics;
- identify and avoid common sources of statistical and backtesting biases;
- conduct reproducible quantitative research in Python.

Prerequisites

Students are expected to possess basic mathematical maturity together with elementary programming concepts. The mathematical and computational tools required throughout the course are progressively introduced and developed when needed.

Course Content

Chapter 1 – Introduction to Quantitative Equity Investing

Lectures: 2 hours

Python Tutorial: –

Topics include

- Active versus passive investing
- Discretionary versus systematic investing
- Alpha, beta and risk premia
- Long-only versus long/short strategies
- Cross-sectional versus time-series investing
- Factor investing
- Quantitative research workflow
- Research reproducibility
- Overview of the course project

Chapter 2 – Financial Data Engineering

Lectures: 4 hours

Python Tutorial: 4 hours

Topics include

- Financial data sources
- Prices, returns and corporate actions
- Fundamental, macroeconomic and alternative data
- Point-in-time databases
- Data lakes and research databases
- Merging heterogeneous datasets
- Missing data treatment
- Data validation
- Data versioning
- Survivorship bias
- Look-ahead bias
- Selection bias
- Data snooping bias
- Publication lag
- Corporate action adjustments

- Benchmark construction biases
- Common backtesting pitfalls

Python tutorial

- Building a research database
- Cleaning financial datasets
- Merging multiple data sources
- Implementing a reproducible data pipeline

Chapter 3 – Data Cleaning and Feature Engineering

Lectures: 4 hours

Python Tutorial: 4 hours

Topics include

- Return definitions
- Forward returns
- Rolling and expanding windows
- Volatility estimation
- Lagged variables
- Feature scaling
- Standardization
- Cross-sectional normalization
- Industry neutralization
- Winsorization
- Clipping
- Missing value imputation
- Outlier detection
- Robust statistics
- Feature engineering for financial prediction

Python tutorial

- Building a complete feature engineering pipeline

Chapter 4 – Signal Construction

Lectures: 3 hours

Python Tutorial: 3 hours

Topics include

- Signal construction
- Ranking methodologies
- Quantile and decile portfolios
- Equal-weight and value-weight portfolios
- Information Coefficient
- Rank Information Coefficient
- Monotonicity analysis
- Spread portfolios
- Factor evaluation

Python tutorial

- Construction and evaluation of classical equity factors
- Momentum
- Value
- Size
- Quality

Chapter 5 – Statistical Models for Return Prediction

Lectures: 7 hours

Python Tutorial: 5 hours

All statistical models are derived mathematically before implementation.

Topics include

- Linear regression
- Ordinary Least Squares
- Maximum likelihood interpretation
- Statistical inference
- Bias-variance trade-off
- Ridge regression
- Lasso regression
- Elastic Net
- Regularization

- Feature selection
- Time-series models
- Autoregressive models
- Model diagnostics

Python tutorial

- Implementing prediction models
- Model comparison
- Hyperparameter tuning

Chapter 6 – Machine Learning for Alpha Generation

Lectures: 4 hours

Python Tutorial: 4 hours

Topics include

- Decision trees
- Random forests
- Gradient boosting
- Feature importance
- Model interpretability
- Cross-validation
- Walk-forward cross-validation
- Expanding and rolling windows
- Hyperparameter optimisation
- Preventing overfitting

Python tutorial

- Training and comparing machine learning models
- Walk-forward evaluation

Chapter 7 – Portfolio Construction

Lectures: 4 hours

Python Tutorial: 2 hours

Topics include

- From signals to portfolio weights
- Equal-weight portfolios
- Long-only portfolios
- Long-short portfolios
- Dollar neutrality
- Market neutrality
- Sector neutrality
- Position sizing
- Turnover
- Transaction costs
- Portfolio constraints
- Capacity considerations

Python tutorial

- Building investable portfolios

Chapter 8 – Portfolio Optimisation

Lectures: 4 hours

Python Tutorial: 2 hours

Topics include

- Mean-Variance Optimisation
- Minimum Variance portfolios
- Equal Risk Contribution
- Risk budgeting
- Tracking error optimisation
- Covariance estimation
- Shrinkage estimators
- Constrained optimisation

Python tutorial

- Portfolio optimisation using Python

Chapter 9 – Backtesting Frameworks

Lectures: 2 hours

Python Tutorial: 2 hours

Topics include

- Backtesting architecture
- Rebalancing
- Execution assumptions
- Slippage
- Transaction costs
- Benchmark comparison
- Performance attribution
- Modular research frameworks

Python tutorial

- Building a modular backtesting engine

Chapter 10 – Performance Evaluation

Lectures: 2 hours

Python Tutorial: 2 hours

Topics include

- Annualised return
- Volatility
- Sharpe ratio
- Information ratio
- Sortino ratio
- Maximum drawdown
- Calmar ratio
- Alpha and beta
- Factor attribution
- Turnover analysis
- Robustness analysis
- Sub-period analysis

Python tutorial

- Building a professional performance report

Final Project

Students work in teams to develop an original quantitative equity investment strategy.

The project includes

1. data acquisition,
2. feature engineering,
3. signal construction,
4. predictive modelling,
5. portfolio construction,
6. backtesting,
7. performance evaluation,
8. presentation of the research findings.

Course Organisation

Lectures	36 hours
Python tutorials	28 hours
Final project	10–15 hours
Total workload	approximately 75–80 hours